Spain

Electricity price forecasting under 2030 energy scenarios

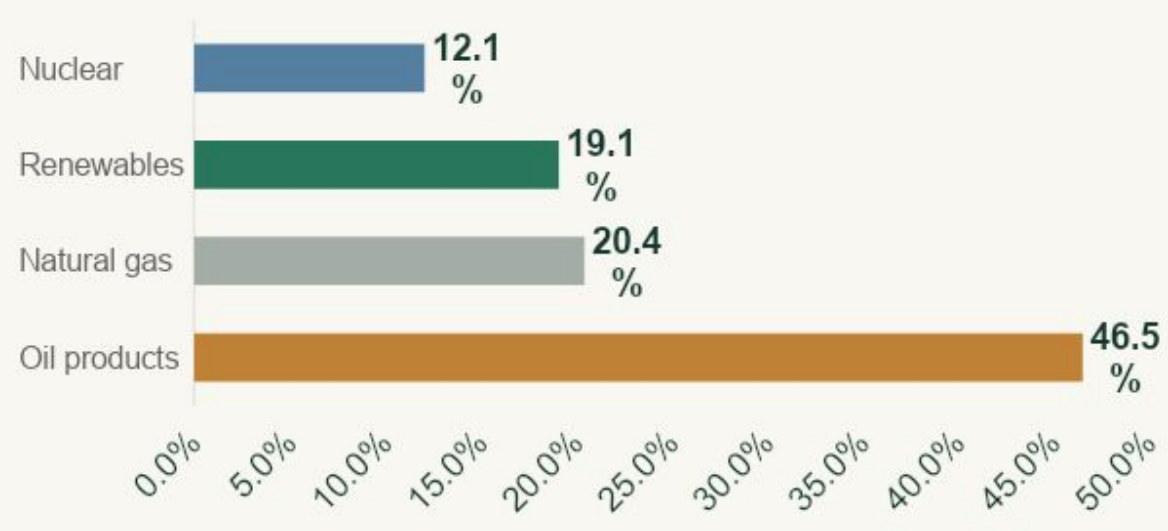
MJ2507 AI Applications in Sustainable Energy Engineering
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Project aim

Use machine learning and Spain’s energy plan to explore how future electricity prices could change.

Spain’s overall energy mix

Primary energy consumption, 2024 snapshot. Includes energy used across the economy.



Fossil fuels still play a large role

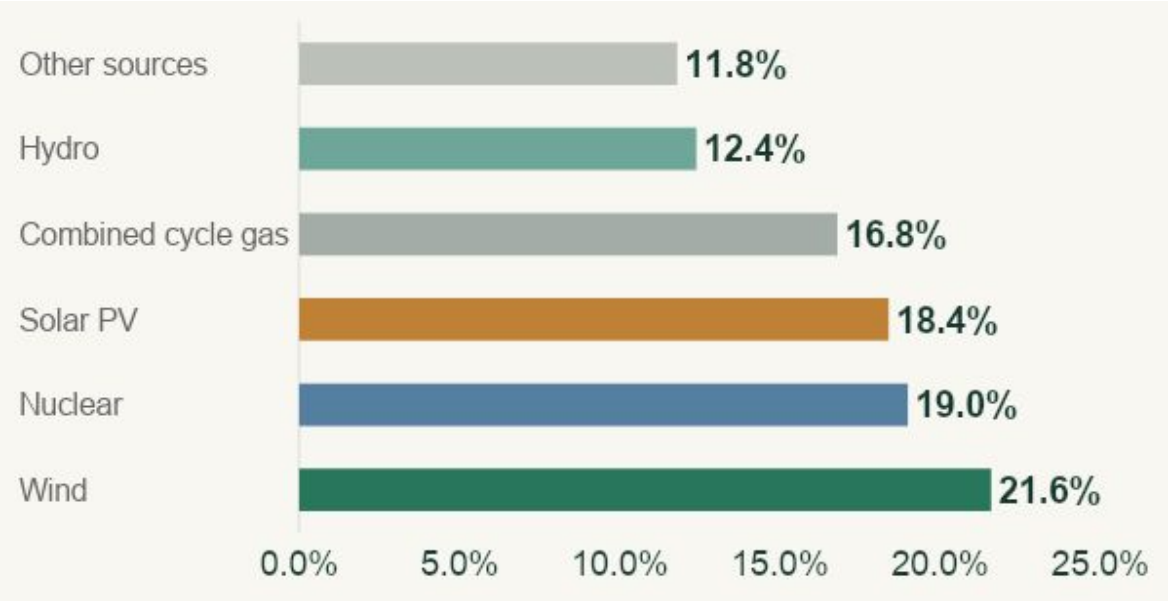
Oil and gas account for about 67% of primary energy consumption.

Selected sources. The remaining balance includes coal, waste and net electricity trade.

Source: MITECO, Balance Energético de España 2024, pp. 3–4.

Spain’s electricity mix in 2025

Share of national electricity generation, excluding estimated self-consumption.



55.5%

renewable electricity

56.6% when estimated self-consumption is included.

Source: Red Eléctrica, Spanish Electricity System 2025, published March 2026.

Spain’s 2030 energy and climate plan

Updated PNIEC 2023–2030, approved in September 2024.

Renewable electricity generation: **81%**

Renewables in final energy consumption: **48%**

Electricity demand: projected growth of 34% from 2019 to 2030

The plan combines renewable expansion with greater use of electricity across the economy.

Source: MITECO, PNIEC update, 24 September 2024.

Renewable capacity and storage targets

Planned installed power in 2030, measured in gigawatts (GW).



3 GW of the wind target comes from offshore (wind turbines at sea)

19 GW of the wind target comes self-consumption (e.g. solar panels on homes

GW describes power capacity. Storage shifts electricity between hours.

Source: MITECO, PNIEC update, 24 September 2024.

How Spain plans to reach 2030

Policy measures supporting the renewable targets and growth in electricity demand.

Renewable generation	Expand solar and wind, including self-consumption.
Electrification	Increase electricity use in transport, buildings and industry.
Storage and flexible demand	Shift electricity use and supply across the day.
Electricity networks	Strengthen grids to connect generation and new demand.

Project relevance: the PNIEC provides a basis for future demand and generation scenarios.

Sources: MITECO, updated PNIEC 2023–2030 and approval summary.

Nuclear phase-out: basis for the scenarios

Seven reactors with 7.1 GW in total, supply (=~19% of Spain's electricity, are being planned to be closed by 2035 under current policy.

Reactor	Net capacity	Closure: 2019 plan	Closure: now
Almaraz I	1,011 MW	2027	**June 2030 ***
Almaraz II	1,006 MW	2028	**June 2030 ***
Ascó I	995 MW	2030	2030
Cofrentes	1,064 MW	2030	2030
Ascó II	997 MW	2032	2032 **
Vandellós II	1,047 MW	2035	Feb 2035
Trillo	1,003 MW	2035	May 2035
Total	**7,123 MW**		

* Extended by the government in Aug 2026. ** Licence runs to 2031; the phase-out plan says 2032.

Sources: World Nuclear Association, Spain country profile (Aug 2026)
World Nuclear News and Euronews, 14 Aug 2026
Red Eléctrica 2025.

What changed in 2026

Almaraz got a limited extension to June 2030 because of gas price volatility. The government says the 2035 end date for the whole fleet is unchanged.

Why it matters for price: nuclear is firm baseload. Removing it leaves gas or storage to cover low-wind nights, which the Sweden-based research paper found raises prices and volatility.

Scenario Design: Spain's Future Electricity Mix

Comparison of three future energy scenarios for Spain's power sector development.

	Scenario 1: Accelerated Transition	Scenario 2: Constant Thermal Backup	Scenario 3: Dissociated Electro-Intensive
Electricity Demand	358 TWh (by 2030)	Moderate / Stagnant growth	> 400 TWh (AI & Green H2 driven)
Renewables & Mix	81% renewable share (160 GW)	High solar curtailment (>15%)	Private PPAs monopolize green energy
Nuclear & Backup	Nuclear phase-out starts in 2030	Nuclear fleet extended (7 GW baseload)	High reliance on foreign nuclear imports

Sources: MITECO, PNIEC 2023–2030, deep research documentation.

Suggested inputs

To achieve hourly day-ahead price, Spanish bidding zone (€/MWh) (Based on Table 7 of the paper)

Input	Resolution	Source	Why
Calendar: hour, weekday/weekend, month, holidays	Hourly	Derived	Daily, weekly and seasonal price pattern
Load forecast	Hourly	ENTSO-E	Strongest driver in the paper's sensitivity test
Wind and solar forecast	Hourly	ENTSO-E / renewables.ninja	Push price down; solar creates the midday dip
Nuclear generation	Hourly	ENTSO-E	Baseload; main scenario lever
Hydro reservoir level	Weekly	ENTSO-E	Flexible supply, varies with rainfall
Gas (MIBGAS/TTF) and CO2 (EUA) price	Daily	MIBGAS / EEX	Gas plants often set the marginal price in Spain
Net imports (France, Portugal)	Hourly	ENTSO-E	Coupled market; a limitation the paper names
Lagged price: P(t−24), P(t−48), P(t−168), day averages	Hourly	ENTSO-E	Largest accuracy gain in the paper

Sources: Sridhar et al. (2022), Table 7 and Table 9; ENTSO-E Transparency Platform; renewables.ninja; MIBGAS; EEX.

Model setup

Suggested steps:

1. Data: hourly data for 2023–2025. 2022 is left out because the gas crisis made prices abnormal

2. Clean: fill missing hours if applicable

3. Features: put the inputs on similar scales, add past prices

4. Split: train on years 2023-24, pick settings using the first half of 2025, and test on the last week(s) of 2025. Split by time so the model never trains on data from after the period it's tested on.

5. Tune: try different model settings and keep the ones with the lowest validation error.

6. Evaluate: forecast the test week. Check MAE (mean absolute error) and RMSE (root mean squared error)

First round: what we expect

Add hypothesis based on the research paper Jagruti sent out